

Complementary Stability and Loop Shaping for Improved Human–Robot Interaction

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Abstract—Robots intended for high-force interaction with humans face particular challenges to achieve performance and stability. They require low and tunable endpoint impedance as well as high force capacity, and demand actuators with low intrinsic impedance, the ability to exhibit high impedance (relative to the human subject), and a high ratio of force to weight. Force-feedback control can be used to improve actuator performance, but causes well-known interaction stability problems. This paper presents a novel method to design actuator controllers for physically interactive machines. A loop-shaping design method is developed from a study of fundamental differences between interaction control and the more common servo problem. This approach addresses the interaction problem by redefining stability and performance, using a computational approach to search parameter spaces and displaying variations in performance as control parameters are adjusted. A measure of *complementary stability* is introduced, and the coupled stability problem is transformed to a robust stability problem using limited knowledge of the environment dynamics (in this case, the human). Design examples show that this new measure improves performance beyond the current best-practice stability constraint (passivity). The controller was implemented on an interactive robot, verifying stability and performance. Testing showed that the new controller out-performed a state-of-the-art controller on the same system.

Index Terms—Actuators, control systems, force control, haptics, man–machine systems, robots.

I. INTRODUCTION AND BACKGROUND

INTIMATE physical interaction and cooperation with humans is an exciting and challenging frontier of robotics. Robotic therapy is a rapidly emerging application which has already proven effective even in limited forms [1], [2]. Robots capable of safe, controllable physical interaction enable this application, as well as haptic devices [3], teleoperated master–slave systems [4], [5], and advanced applications such

as fully programmable exercise machines and cooperative assembly systems. However, in developing therapy robot modules to treat more functions and limbs [6], [7], we have observed that actuator technology is the most prominent limiting factor. Sensor technology [8] has improved dramatically in recent decades (partly enabled by microelectromechanical systems (MEMS) technology), computational capacity continues to grow following Moore's Law [9], but advances in actuator technology have lagged behind. This limitation is particularly acute as we consider *high-force haptics*, machines capable of generating forces up to a human's body weight while providing a gentle, compliant behavior. This requires actuators with low and tunable endpoint mechanical impedance as well as high *force density* (the ratio of force capacity to weight), but available technologies (e.g., electromagnetic, hydraulic, pneumatic) generally excel in at most one of these capacities. In this paper, we consider the design of feedback-control strategies to improve and/or sculpt actuator behavior. We expose challenges of the interaction control problem that have hampered design of effective controllers, and propose a novel design method to shape controller dynamics for improved performance and stability.

We define *physical interaction* between a controlled system and another object to occur when the static or dynamic properties of the system formed by coupling the two together differ significantly from those of the controlled system in isolation. Interacting systems often exchange significant energy, but energy transfer is not necessary to create interaction; for example, a robot pushing on a completely rigid wall transfers no net energy, but is dynamically quite different from the same robot in isolation. Interaction can alter both performance and stability of the controlled system; that is, a system that is stable and satisfies performance objectives in isolation may be unstable or perform inadequately when coupled to another system [10].

Although interaction control has been investigated for several decades, the available design tools remain substantially more cumbersome and less effective than those used for classical servo design. Most controllers for interacting systems either regulate force directly or regulate static and dynamic behavior at the *interaction port*, the location of interaction. Whitney first introduced control based on measured force signals in the late 1970s [11]. Work on direct force control spans a wide range of possible forms, including proportional [12], pure integral (I) [13], [14], proportional-integral (PI) [15], proportional-derivative [16], model-based control [15], and controllers with interior motion-control loops [17]. However, a widely accepted framework to determine the tuning of gains or the placement of controller poles and zeros is lacking. Most prior

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literature in force control argues the merits of one controller form or another without providing a systematic framework to guide controller design.

A generally more robust approach to interaction control regulates dynamic behavior at the interaction port. While conventional motion and force control regulate variables that depend on both interacting systems, this approach regulates properties that depend solely on the controlled system, independent of the environment. One such property is *mechanical impedance*,¹ which determines an output force (torque) time function from an input velocity (angular velocity) time function at the same port. Early work on impedance control [18] assumed that it was possible to achieve arbitrary impedance, and it was postulated that stability could be guaranteed by making a robot emulate a physical system that was itself known to be stable in performing the task (e.g., a damped spring). The impedance could also be sculpted to achieve task performance, for example, maximizing energy transfer [18] or representing “virtual objects” [3], [10].

Unfortunately, it proved to be impossible to achieve truly arbitrary impedance on any real robotic device, regardless of sensor, actuator, or control implementations. In practice, when a controller attempts to emulate dynamics that differ significantly from the intrinsic hardware dynamics, an increased risk of coupled or contact instability arises [19]. This has been explained using the concept of *passivity* (defined in [20]). Colgate proved that when two stable systems with passive port functions are physically coupled together, stability is guaranteed [20]. Because many physical objects are passive (including all combinations of inertias, dampers, springs, and constraints), a stable robot with passive port impedance can couple stably to an extremely large and useful set of environments, making passivity a desirable property. However, Colgate showed that if endpoint force feedback is used to reduce apparent endpoint inertia by more than 50% below its physical value, the system becomes nonpassive. This limitation results from dynamics (regardless of characteristic frequency) between the force sensor and the actuator, holds regardless of implementation, and severely limits the performance of force controllers and most advanced impedance controllers [21]. Discrete control, time delays, actuator and sensor limitations, and unmodeled dynamics can also compromise passivity, making the implementation of passive control on real systems extraordinarily challenging.

Recognizing the fundamental limit on apparent inertia reduction, Newman introduced *natural admittance control* (NAC), an approach that permits reduction of apparent friction independent from inertia, facilitating the design of theoretically passive friction-rejecting interaction controllers [22]. In theory, passivity (and therefore, coupled stability) is ensured if the target inertia is kept above half the physical value, and performance is manipulated by adjusting other target impedance parameters. Because NAC provides a means of addressing both stability and performance in a joint framework, it offers a significant improvement on earlier methods. Yet NAC falls well short of classical tools for servo control design (e.g., root

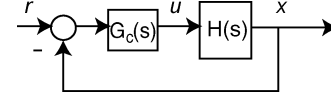


Fig. 1. Block diagram representation of a typical servo-controlled system. $H(s)$ is a scalar-valued complex transfer function of the Laplace variable s that represents the open-loop input-output plant dynamics. $G_c(s)$ is a controller transfer function. A reference command r is compared with the measured output x , and the error is input to the controller. The controller output u serves as the input to the plant.

locus, frequency response design) in the accurate and simultaneous analysis of the effect of control parameters on both stability and performance. Potential threats to stability such as noncollocation, time delay, etc., cannot easily be included; the target inertia is the only measure of passivity or stability and the only parameter that can be intelligently designed for stability margin.

We argue that deficiencies of the tools available for control design are a significant factor limiting the development of physically interactive robotic technology. This paper articulates fundamental differences between the interaction problem and the more common servo control problem as the source of the design difficulties, introduces a method to approximate classical loop-shaping design for interacting systems, and provides a method to incorporate limited knowledge of the environment to improve performance. Section II exposes structural differences between servo and interaction control, and makes the fundamental point that even when analysis is restricted to single-input, single-output (SISO) systems and *even when the robot and environment are both exactly modeled*, servo control design procedures do not translate directly to the interaction problem. In Section III, a procedure is introduced that provides one way to approximate loop-shaping design for interacting systems. Section IV introduces a stability measure for use with the new design method that applies knowledge of the environment dynamics to improve performance. Section V describes the validation on robotic hardware of a controller designed with the new procedure, and compares it with a natural admittance controller. Conclusions are presented in Section VI.

II. CLASSICAL AND INTERACTION CONTROL

Key differences between servo and interaction control require that the two problems be treated with different tools. The differences are most easily demonstrated using scalar functions. SISO controller design, covered in detail in many textbooks (e.g., [23]), consists of shaping controller dynamics to achieve suitable performance and stability margin. Given a realistic representation (often a mathematical model) of sensors, actuators, and system dynamics, including the primary source(s) of instability, familiar tools such as Bode, Nyquist, or root locus analysis can be applied. Given the linear SISO servo system shown in block diagram form in Fig. 1, two key features can be highlighted that enable design with these tools. The first is that although closed-loop stability and performance determine actual system behavior, both can be predicted and characterized using the open-loop transfer function (OLTF) $G_c H$. Bode, Nyquist, and root locus all provide ways to translate properties of the OLTF to the closed-loop system, such that manipulation of the

¹Impedance is generally denoted by the operator Z , and its inverse, the *mechanical admittance*, by the operator Y . Both are examples of port functions, which define behavior at ports of interaction by providing a relationship between input and output energy variables.

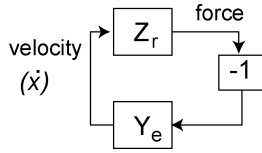


Fig. 2. Block diagram representation of a robot coupled to an environment. Because the robot is represented by an impedance Z_r , the environment is represented by an admittance (Y_e).

former produces predictable effects on the latter. The second key feature is that the controller G_c appears as a direct multiplier of the plant H in the OLTF, such that changes to the controller predictably change the open-loop magnitude and phase, and therefore, stability and performance. In SISO loop-shaping design, performance, stability, and controller complexity are traded off by placing controller poles and zeros and tuning the loop gain. Here we show that these two features are absent from the interaction problem, and that controller design is therefore hindered.

A. Limits of Input–Output Formulation

A first impulse to model an interacting closed-loop system might be to add an “environment” block to the right of Fig. 1, such that the output x serves as the input to the environment. However this would imply an assumption that contact with the environment does not affect the controlled system’s input–output behavior, which directly contradicts the definition of physical interaction provided in Section I. Instead, it is necessary to incorporate any environment dynamics into the model used to derive Fig. 1, generally embedding them in the plant H (an example is carried out in [24]). With this approach, it is possible in theory to derive a stable controller. However, the result would only be valid for a single, precisely modeled environment, and it would be rare to find an environment that could be characterized as accurately and confidently as necessary. With uncertainty in the parameters or particularly the structure of the environment, this approach is even less viable [24]. Direct application of servo techniques to this input–output model of interaction is of little value.

B. Stability and Performance

Input–output block diagram analysis is cumbersome for interaction problems because loading requires independent derivation of a full set of state equations for each environment. Port functions provide a superior means of representing coupled physical systems in block diagram form, because they are based on conjugate power variables, such that their connection implies no assumption about loading [25]. Fig. 2 shows a block diagram model of two coupled single-degree-of-freedom (DOF) systems, a robot and an environment, represented by their port functions. The block diagram representation resembles Fig. 1 in its equivalence to a unity negative feedback system, with the OLTF $Z_r Y_e$ (or $Y_e Z_r$). The system can be analyzed by applying classical SISO tools to this function. However, the use of these methods for design is inhibited by several factors.

Because Fig. 2 provides an exact feedback representation of the interacting system, stability is determined by the poles of

the closed-loop system, or the roots of $(1 + Z_r Y_e)$. Nyquist plots of $Z_r Y_e$ can be used to evaluate stability, and can also be cautiously used to determine some measure of the “distance to instability” or the stability margin, though this measure does *not* translate directly to a scaled controller gain in this case, and must be interpreted carefully, as shown in *C* below.

Performance, on the other hand, requires alternative measures to those used for servo systems. For the system in Fig. 1, performance is measured by the relationship between the input variable r and the output variable x , whether or not a control loop is closed. Those measures do not apply to the system in Fig. 2, partly because the model features no relevant input and output. Further, while measures of interactive performance may vary with system and task, they are best determined by the behavior of the controlled robot alone. When a robot is expected to manage interaction with several different or uncertain environments, to specify performance based on behavior of the coupled system (e.g., natural frequencies, damping) can be impractical; for instance, it is unrealistic (and probably undesirable) to expect comparable dynamic behavior when a robot moves in free space as when it pushes against a rigid constraint, yet both tasks may be needed for some applications. For human-interactive robots, the robot port behavior, colloquially termed “feel,” is usually the best way to represent desired performance [24]. Haptic devices, for example, depend on their ability to emulate appropriate static and dynamic properties to render virtual objects; these properties are quantified by the port function. Thus, quite aside from any stability benefit, the desire to directly regulate port behavior is an important motivation for impedance control.

In terms of the notation of Fig. 2, if performance is measured on the basis of robot behavior alone (and we argue that it usually should be), then it is related to Z_r but not to Y_e ; it is completely independent of the environment. Performance depends exclusively on the portion of the OLTF related to the robot, and performance measures should reflect this. The robot dynamics play a critical role in determining both interaction stability and performance, while the environment dynamics are equally critical to stability but irrelevant to performance. Controller design must consider two quantities: the OLTF $Z_r Y_e$, which determines stability, and the robot impedance Z_r , which exclusively determines performance. This is the first major difference between servo and interaction control.

C. Control and Port Functions

A second major difference between the servo and interaction control transfer function structures hampers predictable manipulation of closed-loop dynamics through the interaction controller. Unlike in the system of Fig. 1, the controller does not appear as a multiplier in the OLTF; in fact, no controller appears directly in the interaction problem as formulated in Fig. 2. The effect of any controller (if there is one) is embedded in Z_r , which also includes the model of the physical robot. If the robot hardware is assumed to be a generic “black box” system, with an actuator driving one side and an interaction port at the other, this can be represented with a “two-port” (a system with two power

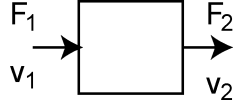


Fig. 3. Generic two-port. The force and velocity at the actuator side, F_1 and v_1 , respectively, and at the environment side, F_2 and v_2 , are shown. In between are arbitrary dynamics. Such a model can be used for any SISO robot.

interfaces that can be connected to other systems), as shown in Fig. 3.

If the robot model is assumed linear, the two-port can be defined by a matrix of transfer functions relating the velocity and force at each port to each other

$$\begin{bmatrix} v_1 \\ F_2 \end{bmatrix} = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix} \begin{bmatrix} F_1 \\ v_2 \end{bmatrix}. \quad (1)$$

Each of the T_{ij} entries are transfer functions. Here the actuator is assumed to be a force source, and the environment interface is in impedance causality. If the port variables v_1 , F_2 , and v_2 are fed back and multiplied by compensator transfer functions G_1 , G_2 , and G_3 , respectively, the resulting control law is

$$F_1 = G_1 v_1 + G_2 F_2 + G_3 v_2. \quad (2)$$

If this is applied to (1), the output impedance at port 2 is

$$Z_r = \frac{F_2}{v_2} = \frac{T_{21}G_3 + T_{22} - T_{11}T_{22}G_1 + T_{12}T_{21}G_1}{1 - T_{21}G_2 - T_{11}G_1}. \quad (3)$$

This shows the relationship between the compensators G_i and the port function. None of the individual compensators (and no combination of them) appears multiplicatively in the port impedance Z_r . No controller enters linearly into either the OLTF $Z_r Y_e$ (the quantity that determines stability), or Z_r (which determines performance), and there is no straightforward way to manipulate the magnitude and phase of either through the controller. A change to the magnitude and phase of G_1 , for instance, transforms the magnitude and phase of Z_r in a complicated way that depends on each of the transfer functions T_{ij} as well as on G_2 and G_3 .

With specific model parameters, stability can be checked using a Nyquist plot, but the contour cannot be reshaped in a straightforward way by altering the compensators. That is why servo tools are ineffective for interaction controller design. This limitation persists even with the assumption of a complete and exact model of the environment Y_e ; it is fundamental to analysis with port functions, which are required to represent interaction in block diagrams. This feature also impacts the interpretation of stability margins: while a significant distance of a coupled system's Nyquist plot from the -1 point may offer some coarse indication of stability robustness, this information *cannot* be translated directly into controller gain or phase, because these quantities may be related in a highly nonlinear way to the port function. Thus, even as an analysis tool, the Nyquist plot has limited value for interactive systems.

Summarizing, in general, SISO tools cannot effectively be applied to design interaction controllers for stability and performance. The elegance that gives these methods their utility for

servo design is lost for interactive design. This is true even for simple SISO robot models, and even in the unlikely case when the environment is well modeled.

III. COMPUTATIONAL LOOP SHAPING

Because classical SISO tools cannot be applied directly to design interaction controllers, alternative methods to facilitate the tradeoff of stability and performance are needed. This section introduces one approach, modeled on loop shaping, that leverages computing power to explore a large set of possible combinations of control parameters. Because there is no obvious way to consider stability and performance as properties of a single quantity (as in the servo problem), they are treated separately before assembling a composite set of results. Measures of stability and performance are established, and constrained optimization is used to search the controller parameter space for solutions that maximize performance while satisfying a stability constraint. The procedure may be repeated to explore changes to the controller structure. The results suggest nonobvious control implementations that can improve performance substantially, as shown in the examples.

Like virtually all controller design and analysis approaches, this method requires a representation of the system to be controlled. It may be arbitrarily complex to satisfy the requirements of a particular problem or application but, at a minimum, it must include the primary source of instability. In the examples below, coupled stability problems originate from attempts to change the robot's apparent port impedance using force feedback. To represent this problem, a model must include at least one resonance between the actuator and the sensor;² a simple inertial model as used in [22] is inadequate. In addition to representing the physical system, a form must be assumed for the controller. This must include at least one variable parameter (and possibly many). It may be helpful for at least one parameter to vary monotonically with performance (e.g., an error-correction gain), but this is not essential. Because the frequency response of the robot port impedance is used for the examples, the robot model and controller are assumed to be linear. Nonlinear models could be used in principle, assuming stability and performance can be adequately quantified, but as yet have not been tested. The computational loop-shaping framework permits various measures of stability. For interaction, stability must be guaranteed when coupled to any expected environment(s). Measures should be selected based on the application, but the stability constraint should not be so conservative as to hinder performance. In *Example 1*, passivity is used, while Section IV describes a new measure that reduces conservatism.

Performance can be quantified by a desired port function; performance error is the deviation from this target function. High admittance has been used as a performance objective for interacting systems, derived from a desire for low contact forces [22]. Here the deviation from the magnitude of the desired impedance (the inverse of admittance) is used; there is no compelling reason to use the phase, and several factors complicate its inclusion [24]. Fig. 4 shows the magnitude and phase of the endpoint impedance for two mechanical systems. The solid trace shows

²Colgate showed that passivity is lost at the same gain level regardless of the complexity of the structural dynamics [21].

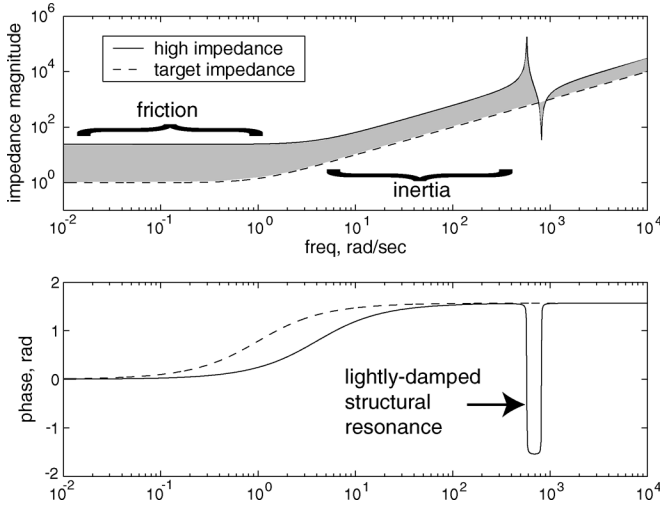


Fig. 4. Port impedance magnitude and phase for a lightly damped mass (target behavior) and a single-resonance system with large mass and damping. The shaded region represents the error.

the impedance of a two-mass, single-resonance system (as in Fig. 6) with large inertia (6 kg total) and damping (24 Ns/m total). The dashed trace shows the impedance of a single, lightly damped mass (1 kg with 1 Ns/m damping), representing a target impedance. On the magnitude plot, the region between the two is shaded to show the error. Below approximately 2 r/s, the difference between the two curves results primarily from the difference in damping. At frequencies between 10 and 200 r/s, inertia dominates, and the difference in offset is due to the difference in apparent inertia. At higher frequencies, the resonance of the two-mass system is apparent. Beyond resonance, a controller can have little or no effect on the endpoint because its action is attenuated by the intermediate dynamics. The phase plot shows the difference in the break frequencies of the systems at low frequencies, and the resonance of the two-mass system around 600 r/s. If the system is controlled to reduce apparent inertia and becomes nonpassive, the phase exceeds $\pi/2$ close to resonance (e.g., see *Example 2*), and the phase error increases, even as the apparent friction and inertia are reduced. This is one reason that phase is not generally useful as a performance measure.

An optimal controller would minimize the deviation from the desired impedance over the relevant frequencies, the shaded area in Fig. 4. A cost function that reflects this for a discrete vector of frequencies between ω_0 and ω_1 is

$$C = \sum_{\omega_0}^{\omega_1} W(\omega) |\log |Z(j\omega)| - \log |Z_{\text{targ}}(j\omega)||. \quad (4)$$

Practice dictates that a log-log scale provides the most useful balance between high- and low-frequency magnitude components, so the frequency vector is logspaced. $Z(j\omega)$ and $Z_{\text{targ}}(j\omega)$ reflect the actual and target port impedance, respectively, and are normalized by 1 Ns/m to render them unitless. $W(x)$ is a scalar weighting function of frequency valued between zero and unity, and is selected by the designer. The weighting function provides a means to emphasize certain frequencies to customize a solution, for example, to reject a resonance. Frequency bounds can also be selected to tailor a solution; here they are chosen to span from several decades

below all resonances to the highest frequencies where control can impact impedance. When $W(\omega) = 1$, C is proportional to a discrete approximation of the shaded area in Fig. 4. The proposed performance measure appears to be adequate for the applications we have investigated. However, it could easily be adapted or augmented to suit alternative applications and systems without compromising the proposed design method. Indeed, any measure that can be modeled and changes with performance is sufficient, though the interpretation of results is facilitated if the cost measure is intuitively (e.g., monotonically) related to a physically meaningful quantity.

The computational design procedure takes the following steps.

- 1) A parameterized robot model is assumed.
- 2) A measure of stability is established, including a model of the environment, if necessary.
- 3) A performance measure is defined, including a vector of relevant frequencies.
- 4) A form for the controller is assumed, with variable parameters P_i . For each parameter, a range of values is chosen.
- 5) For every possible combination of control parameter values, stability is evaluated.
- 6) If a parameter set is unstable, it is eliminated from further consideration.
- 7) If a parameter set is stable, performance is evaluated and stored.
- 8) The stable combination of parameters that yields the lowest cost is selected as the best controller.
- 9) If necessary, the resolution or range of controller parameters P_i is adjusted and the process is repeated.

Depending on the number of parameters varied, the results can be displayed in various ways to view trends due to changing parameter values. If a single parameter P_g (e.g., a gain) is monotonically related to both stability and performance in the range of interest, a stability boundary can be located in the space of P_g for each combination of the remaining parameters. This enables output in the form of one or more “design surfaces” that illustrate the effects of shaping controller dynamics on performance, and affords a definition of a gain-based stability margin. A practical example follows.

Example 1: Loop Shaping for Passive Robot Control: In this example, force feedback is used to reduce the apparent impedance of a single-DOF robot while maintaining passivity of its interaction port. The controller is for a robot module (Fig. 5) used as a portion of a physical therapy robot [26]. The module is backdrivable, but has substantially higher position-dependent Coulomb friction (up to 20 N; target is 2 N) and inertia (6 kg; target is 1 kg) than desired. Force measured by a transducer at the handle of the robot is fed back such that the motor is used to reduce the apparent endpoint impedance. Previous work used proportional force feedback on this robot [26]; this example demonstrates the insights gained by the new method and the performance improvement that results from shaping the controller dynamics.

A linear robot model, depicted in Fig. 6, is used to facilitate the use of the necessary computational tools. A first structural resonant mode and linear damping are modeled; the significant nonlinear Coulomb friction is left unmodeled. The port

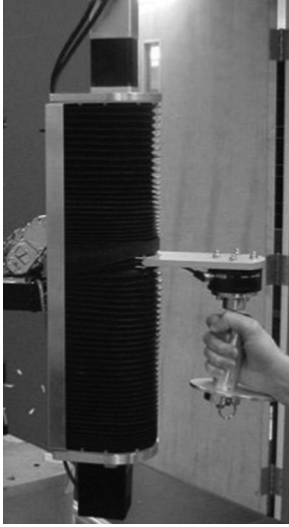


Fig. 5. Screw-driven linear robot module.

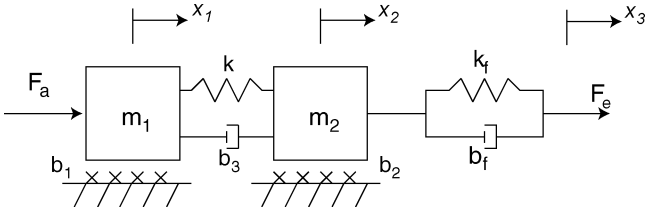


Fig. 6. Single structural resonance robot model with force transducer dynamics. The actuator provides the force F_a . The spring k and damper b_3 represent the first structural resonant mode, while the mass is split between m_1 and m_2 , whose positions are measured by x_1 and x_2 , respectively. b_1 and b_2 represent damping to ground. F_e is the applied force due to interaction with the environment, while k_f and b_f represent the dynamics of the force transducer.

impedance is determined by the dynamic ratio of the endpoint force F_e to velocity $\dot{x}_3$ at the port

$$Z_r = \frac{F_e}{\dot{x}_3}. \quad (5)$$

Once the control law is applied to determine F_a , the port impedance can be derived from equations of motion; the complete port impedance as well as the numerical parameter values used for the example are provided in the Appendix.

For this example, the target impedance (Z_{targ}) is zero at all frequencies; this is equivalent to “pure” force control. This simplifies the controller, as only force feedback is needed; the example is extended to nonzero impedance in [24]. Because the log of zero cannot be computed, (4) must be adjusted for this special case and the alternative measure below is used

$$C_0 = \sum_{\omega_0}^{\omega_1} W(\omega) \log |Z(j\omega)|. \quad (6)$$

Performance is not weighted with frequency, so $W(\omega) = 1$. A logarithmically spaced vector of frequencies ranging from 10^{-2} to 10^3 r/s is used. Theoretically, this cost may be less than zero; however, smaller costs are always preferred. This measure is most useful for ranking candidates rather than as an intrinsically meaningful absolute measure. Impedance plots can be examined

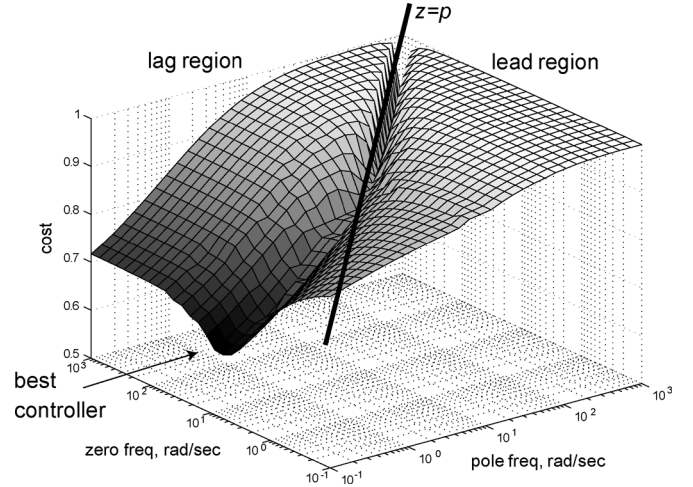


Fig. 7. Normalized cost at stability boundary versus pole and zero location.

to improve understanding of the effects of specific controllers, as illustrated below.

A controller with a single pole and a single zero is used

$$F_a = K_{dc} \frac{(s+z)}{(s+p)} F_e. \quad (7)$$

This controller has three adjustable parameters P_i : the zero frequency z , the pole frequency p , and the DC gain K_{dc} . In the region of interest, K_{dc} is monotonically (though not linearly) related to performance and stability; for fixed z and p , as K_{dc} increases, performance improves while the system becomes “less passive.” This controller form admits lead, lag, and proportional control, depending on the relative values of z and p , permitting the exploration of a range of different controllers with relatively few parameters. Root frequencies were tested from 0.1 r/s, which is well below any significant robot dynamics, to 10^3 r/s, which is above the dominant frequency of the structural dynamics; selection of parameter ranges required no assumptions about the system other than knowledge of the resonant dynamics. To locate the stability boundary for each combination of the values of z and p , the range of K_{dc} was chosen to include both stable and unstable solutions. By exploring controllers of various shapes and parameters, this approach approximates loop-shaping design.

For each set of parameters, the port impedance $Z_r(s)$ is computed, with $s = j\omega$, and evaluated along a vector of frequencies encompassing all relevant dynamics to determine passivity of the interaction port. If the real part of the port impedance is greater than zero at all frequencies

$$\text{Re}(Z_r(j\omega)) > 0 \quad \forall \omega \quad (8)$$

then the port impedance is “positive real” and strictly passive (see [20]). If the real part of the port impedance is less than zero at any frequency, the particular controller renders the system nonpassive, and therefore, this stability measure is not satisfied; that is, there exist passive objects which will destabilize the robot upon contact.

The computational loop-shaping algorithm was used to search a range of values for z , p , and K_{dc} . The results are

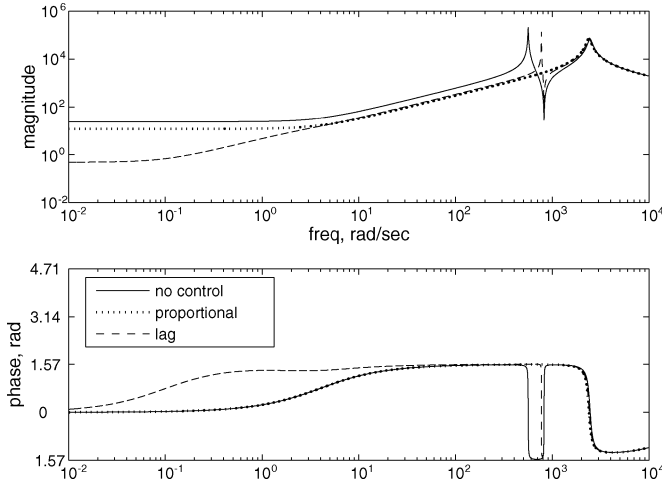


Fig. 8. Lag compensator's effect on robot port impedance.

plotted on the design surface in Fig. 7, which shows the cost with the maximum stable gain K_{dc} for each combination of values for z and p . The cost is normalized such that any cost less than one reduces apparent impedance; where the cost is lower, performance is better. On the diagonal $z = p$, the pole and zero cancel out, yielding proportional control. Performance is best when low-frequency lag compensation is used, with a very low-frequency pole and a higher frequency zero. The placement of the zero has a strong effect on performance. The algorithm dictates that a zero around 6 r/s is optimal (with a normalized cost of 0.72), a far-from-obvious result. The plot in Fig. 7 shows that performance continues to improve as the pole is moved to the lowest tested frequency. Indeed, further testing confirmed that the best performance was obtained when the pole was moved all the way to zero [$p = 0$ in (7)], turning the lag compensator into a PI controller; at this extreme, the cost was 0.63 [24]. Because performance is best with a zero at 6 r/s, PI control is superior to pure I control. In this case, the lead region, where $z < p$, shows little promise.

Fig. 8 shows the port impedance magnitude and phase under the lag controller specified by the loop-shaping algorithm, along with the uncontrolled impedance. Also plotted is a proportional controller with a loop gain of one, which provides a 50% reduction in apparent inertia, the limiting value for passivity. The lag controller is particularly effective at reducing the low-frequency impedance, where friction dominates. The phase stays between $-\pi/2$ and $+\pi/2$, confirming that the interaction port behavior is passive. The apparent inertia at the stability boundary is approximately 3 kg (half the physical value, as expected at the limit for passivity).

The controller boosts the compensation where possible without compromising passivity, while limiting feedback in the frequency range where it is most destabilizing. The algorithm provides a nonobvious frequency-domain “shape” for the controller (through the placement of the control zero) in spite of the nonlinear relationship between the controller dynamics and port impedance. Note that the effect of the controller on the impedance magnitude and phase are not obvious by inspection or by applying classical tools, so the numerical method facilitates the design process.

IV. COMPLEMENTARY STABILITY

To further improve performance, it is necessary to relax passivity and replace it with an alternative measure of interaction stability. While human-interactive robots must accommodate an uncertain environment, the dynamic behavior of that environment (e.g., a human limb) is known to reside within specific limits. The dynamic properties of human arms, for instance, vary considerably between subjects and with muscle activation, but the variation is bounded. The apparent inertia at the hand varies from that of a small child to that of a large adult, but it is never zero and never exceeds a maximum (of several kg). Stiffness is similarly bounded and, perhaps most importantly, an arm can never exhibit certain challenging behaviors such as total physical constraint or extreme static friction. A substantial literature documents dynamic properties of human limbs (e.g., see [27] and [28]).

When passivity is used as a stability objective, the only assumption made about the environment is that it, too, is passive. This is likely sufficient to guarantee coupled stability with humans (though, to date, it has not been conclusively proven that human limbs are passive; see [29] for an argument for treating them as such). However, given the properties of human arms described above, passivity is unnecessarily restrictive. Our experience has shown that some controllers that are known to be nonpassive are adequately stable in clinical rehabilitation tasks [26]. The better defined the conditions for stability, the less conservative controllers can be made, providing latitude to design for better performance. We introduce a new way to define coupled stability, and use it with the loop-shaping method to improve performance.

A controlled robot should be complementary to the environments with which it is designed to interact; that is, it must interact stably with these environments. A controller should provide for this while maximizing performance. We propose the term complementary stability to describe the stability target for interacting machines, where the details of stability are determined by the expected environment(s). Because the dynamics of a particular human limb are almost impossible to know exactly at a given time, an exact model cannot be used for human-interactive robots. However, because all human arms are dynamically similar, the environment can be represented in terms of a nominal port function $Y_n(s)$ and some bounded deviation from this nominal

$$Y(s) = Y_n(s) + W(s)\Delta(s) \quad (9)$$

$\Delta(s)$ defines the possible deviation from the nominal, and is subject to a bound on its magnitude, e.g., in the infinity norm

$$\|\Delta(s)\|_\infty \leq 1. \quad (10)$$

$W(s)$ is a weighting function that shapes the range of possible deviation with frequency. Equation (9) defines a set of dynamic behaviors that includes all expected environments. $Y_n(s)$ and $\Delta(s)$ are chosen based on data or models. The additive perturbation is presented as a representative example, but multiplicative or feedback structures could also be used.

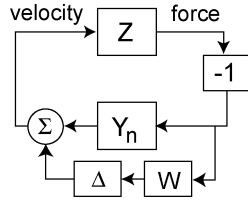


Fig. 9. Block diagram of a robot interacting with an uncertain environment.

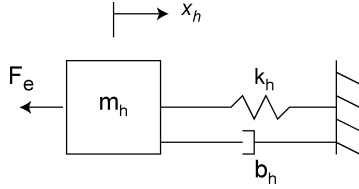


Fig. 10. Second-order environment model.

If the environment is defined by (9), the coupled system can be represented by the block diagram in Fig. 9, which provides the means to evaluate complementary stability for this system.

Definition: A robot with port impedance function Z satisfies the complementary stability condition with a set of environments defined by (9) if the model in Fig. 9 is robustly stable for any $\Delta(s)$ satisfying (10).

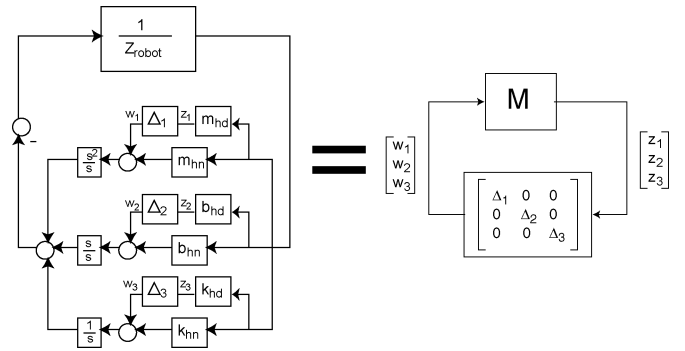
This definition translates coupled stability into a robust stability problem, which can be solved by application of the small gain theorem [30]. Robust stability is equivalent to stability of the robot when coupled to all environments that it is expected to encounter; exactly the stability condition needed. If this condition allows it, nonpassive controllers are permitted; this can improve performance, as shown below.

Example 2: Using Limited Knowledge of Human Dynamics: Example 1 is repeated with complementary stability replacing passivity as the stability measure. The robot model remains as in Fig. 6, and the control law (7) is used along with the performance cost (6). This enables direct comparison of the passive and nonpassive designs.

Complementary stability requires a representation of the environment. Here, a second-order model (shown in Fig. 10) is chosen, capturing inertial (m_h), stiffness (k_h), and dissipative (b_h) characteristics, all of which are important parts of human arm dynamics. Recent work suggests that actual arm endpoint characteristics are close to second-order [27], [28], [31]. Each parameter is defined by a nominal value (m_{hn}, b_{hn}, k_{hn}) and a maximum deviation from that value (m_{hd}, b_{hd}, k_{hd}), such that

$$\begin{aligned} m_h &= m_{hn} + \Delta m_{hd} \\ b_h &= b_{hn} + \Delta b_{hd} \\ k_h &= k_{hn} + \Delta k_{hd}. \end{aligned} \quad (11)$$

Δ is a real number between -1 and $+1$; when stability is achieved for all values in this range, then it is guaranteed for all permitted values of m_h, b_h , and k_h . The values of the parameters in (11) were determined from the literature [27], [28], [31], are given in the Appendix, and include behaviors ranging from primarily inertial to primarily elastic and primarily dissipative, as well as combinations of these behaviors. While parametric


 Fig. 11. Block diagram of robot system interacting with perturbed environment. Expression for M is provided in the Appendix.

uncertainties are used here for their intuitive relationship to physical quantities, the use of the small gain theorem permits a wide range of alternative expressions of uncertainty, including unmodeled dynamics and representations derived from experimental data.

The environment model characterized by Fig. 10 and (11) is most easily represented as an impedance with three perturbation terms. Because the robot impedance is linear and nonzero, it can be inverted and represented as an admittance. A block diagram of the perturbed system is shown in the left panel of Fig. 11, and is equivalent to the system shown in the right panel (M is given in the Appendix). In the latter representation, the robust stability of the system can be determined by the application of the small gain theorem with the *structured singular value* [30]. In this case, the perturbation matrix is diagonal and real-valued; the structured singular value for this case has no closed-form solution and can be computationally burdensome, but it yields a necessary and sufficient condition for coupled stability with all environments in the set described by (11). For each combination of controller values, the algorithm tests stability by computing the appropriate structured singular value μ for M over a set of frequencies where the robot impedance is nonpassive (and therefore coupled stability is at risk); when $\mu > 1$, contact with some environment in the set drives the interface unstable, and the controller is disallowed. For stable solutions, (6) is evaluated to compare performance.

Fig. 12 shows the results of the loop-shaping algorithm using complementary stability. This plot shows the cost (normalized as in Fig. 7) with the maximum gain K_{dc} that meets the complementary stability requirement for each combination of values for z and p . As in Example 1, performance is best when low-frequency lag compensation is used (marked **A** in Fig. 12); the normalized cost is 0.50 for the limiting plotted value (again, the extreme case of PI control offers a small additional improvement [24]). A lead controller at high frequencies (labeled **B** in Fig. 12) is almost as effective at lowering cost. The best performance in this region has a normalized cost of 0.59, when $z = 280$ and $p = 730$ r/s. The design algorithm identifies two distinct (and nonintuitive) control forms that each substantially improve performance; both offer significant improvement over passive control.

The impedance plots in Fig. 13 contrast the controllers. The best proportional, lag, and lead controllers selected by

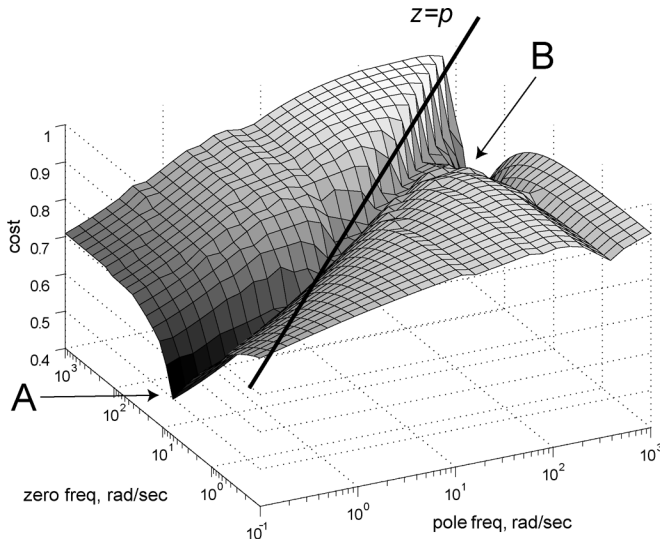


Fig. 12. Normalized cost at stability boundary versus pole and zero frequency.

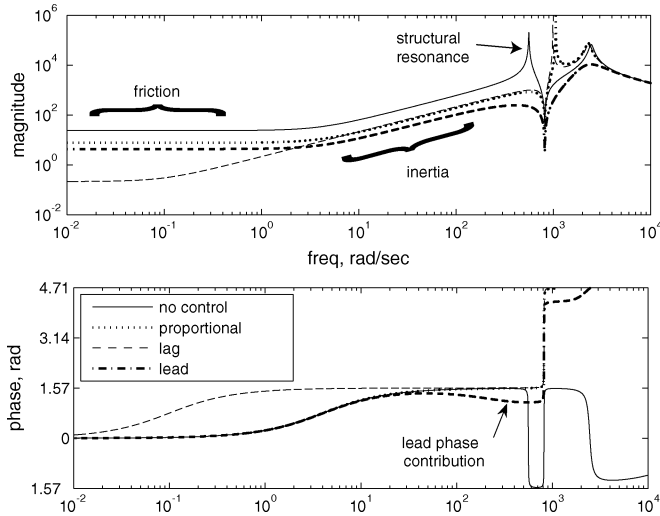


Fig. 13. Port impedance magnitude and phase under no control, proportional, lag, and lead control.

the method are shown along with the uncontrolled system impedance.

All three controllers are nonpassive, as evidenced by the phase exceeding $\pi/2$ in the frequency range of the structural dynamic behavior (around 10^3 r/s), but they ensure coupled stability specifically because knowledge of the environment has been used for design. The magnitude plot shows that the lag and lead controllers provide very different benefits compared with proportional control: the former primarily boosts the gain at low frequencies to offset friction, while the latter reduces the impedance (inertia as well as friction) more equally at all frequencies. The difference can be understood by considering the phase plot. The lag compensator takes effect at low frequencies, but the zero and gain are placed such that the phase stays below $\pi/2$ until close to the structural resonance, where it then tracks the phase of the system with proportional control. The lead compensator creates a substantial dip in phase just below resonance, reducing the destabilizing phase and permitting

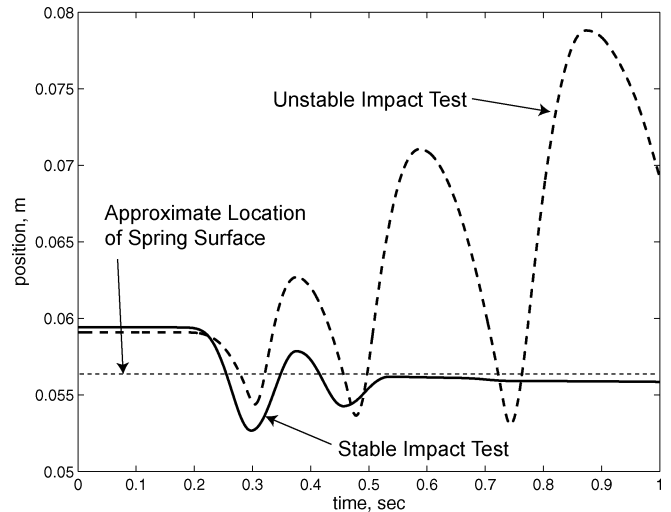


Fig. 14. Examples of stable and unstable trajectories from impact tests.

a higher gain value across all frequencies. The nonintuitive nature of this controller's effect underscores the need for a method like the one described here. The lag and lead controllers are separated by almost two decades in frequency, and can be combined into a lag-lead controller to reap the benefits of both [24].

Compared with *Example 1*, the new stability measure reduces the minimum cost from 0.72 to 0.50. Loop shaping improves performance, and complementary stability parlays knowledge of the environment into further performance gains. The latter result is robust to realistic environmental uncertainty; an exact model is not required to reap benefits. Though not demonstrated here, this approach also provides a means of controller design when the environment may be nonpassive.

V. IMPLEMENTATION AND COMPARISON WITH ALTERNATIVES

The controller from *Example 2* was tested on the robot module in Fig. 5 using a PC-based control system (QNX O/S, 1000 samples/s). Because of bandwidth limitations in the force measurement, high-frequency lead compensation could not be implemented as designed. A lag compensator with a break frequency of 6 r/s (selected by the design method) was tested.

The coupled stability properties of the system were tested by contact with springs, a hard plastic block, and human arms. Stiff springs are usually the most destabilizing environment for systems that use force feedback [20]. A "contact test" and an "impact test" were performed with six different springs with stiffness ranging from 260 to 57 000 N/m, and with the plastic block, with stiffness of approximately 10^6 N/m. In the contact test, the robot handle was brought by hand into resting contact with the spring and released. If oscillations grew, this test was judged unstable. The more challenging impact test required that the robot fall a short distance (1–2 cm) to contact the environment and withstand the impact. If oscillations abated and the system settled to rest, this test was judged stable, while if oscillations grew, it was judged unstable. Fig. 14 shows examples of stable and unstable impact tests. Finally, each controller was tested by vigorous grasp with a human arm: a handle on the robot module was gripped tightly and the arm and hand muscles co-contracted

TABLE I
 COUPLED STABILITY PREDICTIONS AND TEST RESULTS, LAG CONTROL

Increasing Environment Stiffness (N/m) →									
	none	263	976	1.78e3	2.65e3	1.64e4	5.71e4	block 1.0e6	arm
Kdc=850							PXO	++	+
Kdc=2000	P					X	O	++	+
Kdc=3000	P				X		O	++	+
Kdc=5000	P				X		O	+-	-

Controller specified by algorithm
 Explicitly part of design set
 X Maximum stiffness stable in impact test
 O Maximum stiffness stable in contact
 P Predicted maximum stable stiffness

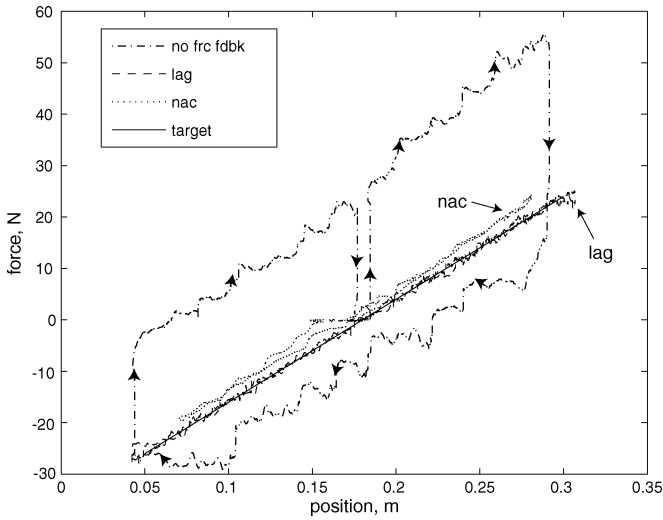


Fig. 15. Virtual spring implemented with controller with no force feedback, lag controller, and natural admittance controller. The direction of travel is shown by the arrows on the uncompensated system curve.

maximally. The subject varied arm geometry and effort level to test as wide a range of stiffness, damping, and inertia as possible. If any vibration or oscillation could be induced, the controller was marked as failing this test.

Table I shows composite results from coupled stability testing, for the lag compensator with several different DC gain parameter values. The columns marked with stiffness values only show the maximum stable spring stiffness for each gain level tested (separated by rows), as well as the maximum stable stiffness predicted by the design model. The next two columns show the results of testing with the plastic block and the human arm; “+” indicates a stable coupling and “-” indicates instability. In the “block” column, the left mark is for the contact test and the right for the impact test. The design set, which includes only the lowest spring stiffness, is also highlighted. The controller proves quite robustly stable under these test conditions; in fact, only the highest gain level tested produced any problem interacting with human arms. The system has even better stability properties than predicted, enabling better performance through a higher DC gain.

As predicted, the lag controller is particularly effective at rejecting friction, thanks to its high gain at low frequencies. This can be seen in Fig. 15, which shows the force versus displacement curve for a virtual spring implemented on the

robot module. These curves were produced by slowly moving the robot by hand against the controllers. The uncompensated (dash-dot) curve shows the virtual stiffness implemented with proportional position feedback alone, and thus reflects all position-dependent friction intrinsic to the hardware. Coulomb friction appears as the vertical segments of the curve, which occur when starting, stopping, or when direction is reversed. With the new compensator (dashed line), the low-frequency behavior tracks the desired curve very closely. The Coulomb friction is reduced from 10–20 N to less than 0.5 N, a reduction of at least 95%. Inertia, tested by initial condition response against a virtual spring of known stiffness, is just 1.2 kg when $K_{dc} = 3000$, representing a reduction of approximately 80% from the physical value (6 kg) and 60% from the theoretically best passive controller (3 kg). This impedance reduction is impressive in its own right, but even more so when considered in light of the robust stability properties of this system. With apparent friction reduced by 95% and inertia by 80%, the system reliably remains stable despite a substantial impact with the plastic block; a difficult task for any stiff robot, particularly one using force feedback.

In all cases, the model is more conservative than the test; that is, the model predicts instability for cases where the test system remains stable. This may result from the fact that the system’s Coulomb friction, potentially a large energy sink, is unmodeled. This suggests that progress can in some cases be made using simplified, linear models, even when robot behavior is significantly nonlinear, and serves as an encouraging endorsement of this approach. While an exact model may be preferred, a simplified linear model permits the use of a vastly greater suite of analytical tools and is often more realistically attainable on real hardware.

The controller described above was compared with NAC, the current “best practice” for interaction control [22]. The NAC structure uses both motion and force feedback; target dynamic properties are specified, the interaction force is measured, and the motion feedback controller is used to servo the system to the response specified by a target admittance. To design a NAC, the module in Fig. 5 is modeled as a damped mass. The controller uses both motion and endpoint force feedback

$$F_a = G_v v + G_f F_e. \quad (12)$$

v is the velocity of the mass, and F_a and F_e are again the actuator and interaction force, respectively. G_v and G_f are controller

TABLE II
COUPLED STABILITY PREDICTIONS AND TEST RESULTS, NAC

	Environment Stiffness, N/m								
	none	263	976	1.78e3	2.65e3	1.64e4	5.71e4	block 1.0e6	
0	XO							--	+
0.1	XO							--	+
0.3	X	O						--	+
0.6	X	O						--	+
1		XO						--	+
2		XO						--	+
3		X	O					--	+
4			XO					--	+
5			XO					--	+
7				X	O			--	+
10					XO			--	+
15					XO			--	+
20					XO			--	+
25					XO			--	+
35						XO		--	+
50						XO		--	+

transfer functions. To derive the controller, the port admittance of the system, with (12) applied as the control law, is derived and set equal to a desired admittance function with stiffness k_{des} , damping b_{des} , and mass m_{des}

$$\frac{G_f + 1}{ms + b - G_v} = \frac{1}{m_{des}s + b_{des} + k_{des}/s}. \quad (13)$$

Proportional velocity feedback is used so that $G_v = K_v$, and the compensator for force feedback can be derived from (13). For passivity, [22] recommends selecting m_{des} at no less than $(1/2)m$. While NAC was introduced as a means of providing passivity, it can be used to make nonpassive controllers with better performance if the target mass is further reduced [32].

NAC was implemented with a range of target parameters, and stability and performance experiments were repeated. A sample of the results is shown for $m_{des} = 2$ kg and variable b_{des} in Table II, which uses the same form as Table I. In general, the stability properties of nonpassive controllers designed with the NAC method were inferior to those that employed lag force-feedback control. Only when the target inertia m_{des} was set at 6 kg (i.e., no reduction of apparent inertia) and the target damping at 50 Ns/m was the NAC-controlled module as stable in contacting all three types of environments (springs, block, and arm) as the lag controller with $K_{dc} = 3000$. In this case, however, the extremely high viscous friction and inertia make performance far inferior. Complete results are in [24].

NAC is relatively effective in making the system exhibit a linear dynamic response, because servoing along an actual linear system trajectory is a specific objective. However, because velocity is fed back rather than position, it is subject to accumulated errors in implementing desired stiffness, causing a drift phenomenon. A hand-actuated force-position curve is shown in Fig. 15 for comparison with the other controllers. The lag controller's use of low-frequency gain (approximating integral action) makes endpoint behavior more noticeably nonlinear than with NAC, but it effectively suppresses friction nonetheless, and creates a very accurate virtual spring. Both controllers have performance strengths, but in this case, the

lag controller proved to be more robust to interaction with the target environment.

VI. CONCLUSIONS

Despite decades of effort, interaction control remains a challenging problem, and the design of interaction controllers lacks the unified, effective approach enjoyed in servo design. We believe that this results, at least in part, from the underappreciated structural differences between the two problems, and the continued attempts to apply servo design techniques to interacting systems. The approach described here seeks to bridge the gap between the two control problems; the design "surfaces" of Figs. 7 and 12 are the output of this method, and provide an approximate equivalent to the insight available intuitively from servo loop shaping. The examples show that, not surprisingly, there are significant benefits to be gained from shaping the dynamics of feedback controllers, and also that greater benefits can be achieved by relaxing passivity.

The examples presented here used a primitive optimization technique. Advanced tools could almost certainly improve efficiency. Nevertheless, displaying the entire parameter space to see trends (as in Figs. 7 and 12) rather than finding only one best-performing solution provides valuable insight.

To date, the method has been applied to single-DOF systems, where it has paid immediate dividends for actuator control. Extension to multiple-DOF robots would be useful; however, it is important not to understate the value of the single-DOF case. The most significant limitation in the development of more advanced interactive robots comes from the drive system; achieving both high force density and low mechanical impedance is a persistent, vexing problem [24]. Single-DOF control methods that can significantly improve actuators will enable a significant technological advance and facilitate increased implementation of interactive robots. Application to multi-DOF robots represents an important area for continued study. Although extension is nontrivial, at least two different approaches show promise. One option is to place force sensors on each actuator and design controllers with an adapted version of the method presented here. The principal challenge is that the "environment" with which each actuator interacts could vary with robot configuration. For this reason, this approach may be particularly fruitful for robots that typically operate in only a small fraction of their total configuration space (e.g., MIT-MANUS [1]), such that changes in dynamics with configuration are minimal. A second approach would use endpoint force feedback and shape force compensation directly at the robot level. This approach should be feasible, but will require addressing several issues, including modeling the primary robot resonances (which are likely to depend on configuration) and defining performance at a multi-DOF interaction port [33]. Either approach might benefit from a gain-scheduling controller that adjusts the dynamics of force compensation with configuration.

APPENDIX

If the generic control law

$$F_a = G_1(s)x_1 + G_2(s)x_2 + G_f(s)F_e \quad (A-1)$$

TABLE III
 PARAMETER VALUES FOR ROBOT AND ENVIRONMENT MODELS

Parameter	Value	Units	Parameter	Value	Units
m_l	3	kg	m_{hm}	2.1	kg
m_2	3	kg	m_{hd}	2.0	kg
b_l	12	Ns/m	b_{hm}	21	Ns/m
b_2	12	Ns/m	b_{hd}	20	Ns/m
b_3	1	Ns/m	k_{hm}	201	N/m
k	10^6	N/m	k_{hd}	200	N/m
k_f	1.7×10^7	N/m			
b_f	700	Ns/m			

where $G_1(s)$, $G_2(s)$, and $G_f(s)$ are controller transfer functions, is applied to the model in Fig. 6, the impedance of the system upstream of the force sensor dynamics $F_e/\dot{x}_2$ is the following:

$$\begin{aligned} \frac{Z_n(s)}{Z_d(s)} = & \frac{m_1 m_2 s^4 + (m_2 b_3 + m_2 b_1 + m_1 b_3 + m_1 b_2) s^3}{m_1 s^3 + (b_1 + b_3) s^2 + k s - G_1 s + G_f s (b_3 s + k)} \\ & + \frac{(b_2 b_3 + b_1 b_3 + b_1 b_2 + m_1 k + m_2 k) s^2}{m_1 s^3 + (b_1 + b_3) s^2 + k s - G_1 s + G_f s (b_3 s + k)} \\ & + \frac{(k b_2 + k b_1) s - G_1 m_2 s^2 - G_1 (b_3 + b_2) s}{m_1 s^3 + (b_1 + b_3) s^2 + k s - G_1 s + G_f s (b_3 s + k)} \\ & - \frac{G_1 k + G_2 b_3 s + G_2 k}{m_1 s^3 + (b_1 + b_3) s^2 + k s - G_1 s + G_f s (b_3 s + k)}. \end{aligned} \quad (A-2)$$

When the force transducer dynamics are included, the robot port impedance becomes

$$Z_r(s) = \frac{(b_f s + k_f) Z_n(s)}{(b_f s + k_f) Z_d(s) + s Z_n(s)}. \quad (A-3)$$

The port impedance with a particular controller such as (7) can be determined by substituting into these equations. If the environment is as shown in Fig. 10 and described by (11), then the expression for M in Fig. 11 is

$$M = \frac{\frac{1}{Z_r}}{1 + \frac{m_{hn} s^2 + b_{hn} s + k_{hn}}{s Z_r}} \begin{bmatrix} m_{hd} s & m_{hd} & \frac{m_{hd}}{s} \\ b_{hd} s & b_{hd} & \frac{b_{hd}}{s} \\ k_{hd} s & k_{hd} & \frac{k_{hd}}{s} \end{bmatrix}. \quad (A-4)$$

Parameter values for both examples are provided in Table III.

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